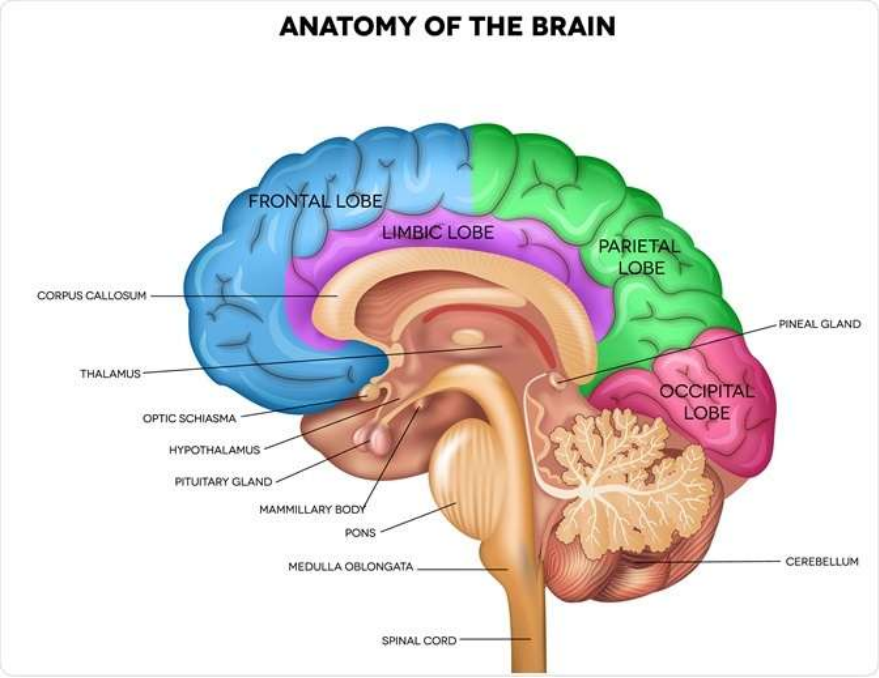
. Introduction

Intelligence refers to the ability to learn, understand, adapt, solve problems, and apply knowledge. Intelligence exists in different forms:

- **Human Intelligence** – Natural intelligence of humans.
- **Animal Intelligence** – Cognitive abilities of non-human animals.
- **Machine Intelligence** – Artificial intelligence developed in computers and machines.

This assignment compares these three forms using charts and diagrams.

2. Human Intelligence



4

Definition

Human intelligence is the natural cognitive ability of humans to think, reason, learn, and solve problems.

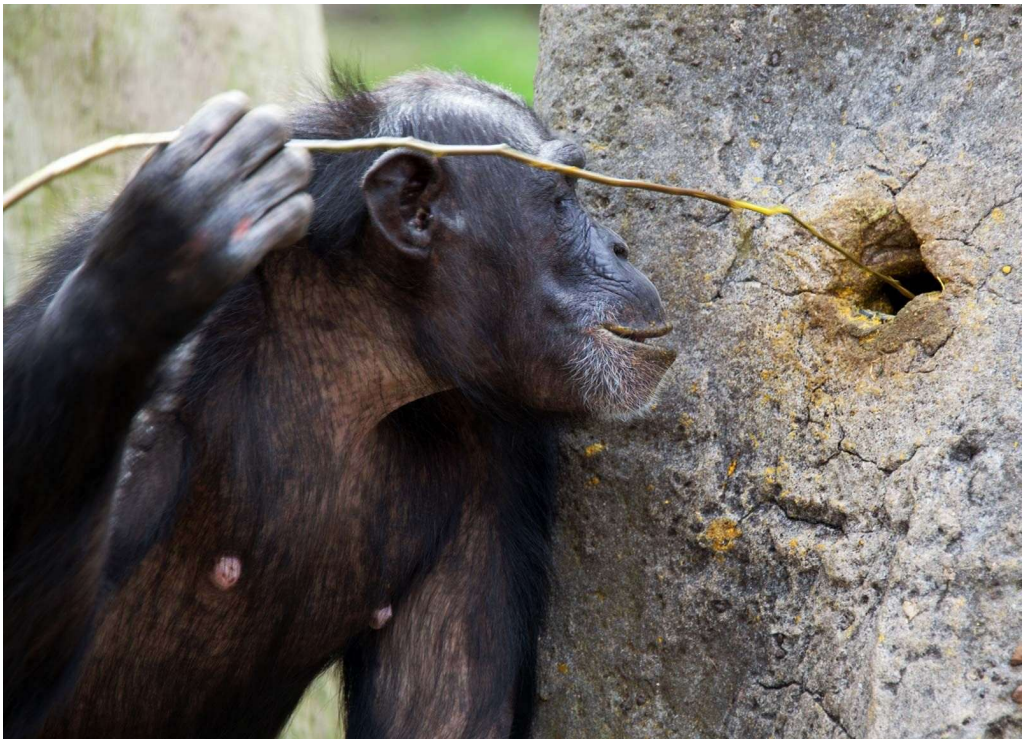
Characteristics

- Logical reasoning
- Emotional understanding (Emotional Intelligence)
- Creativity and imagination
- Self-awareness
- Moral judgment
- Language and communication

Example

A student solving a complex mathematical problem using reasoning and creativity.

3. Animal Intelligence



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Definition

Animal intelligence refers to the cognitive abilities shown by animals such as learning, memory, and problem-solving.

Characteristics

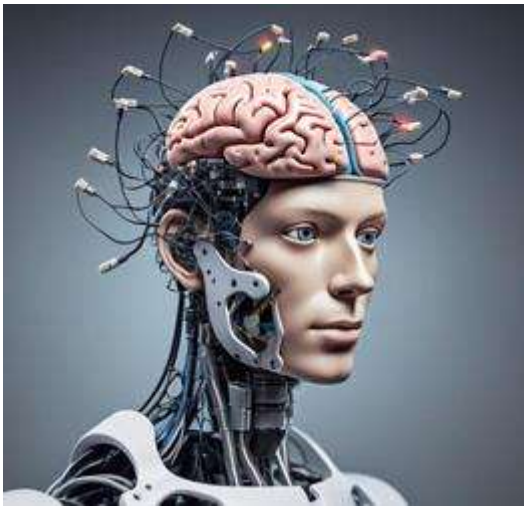
- Instinct-based behavior

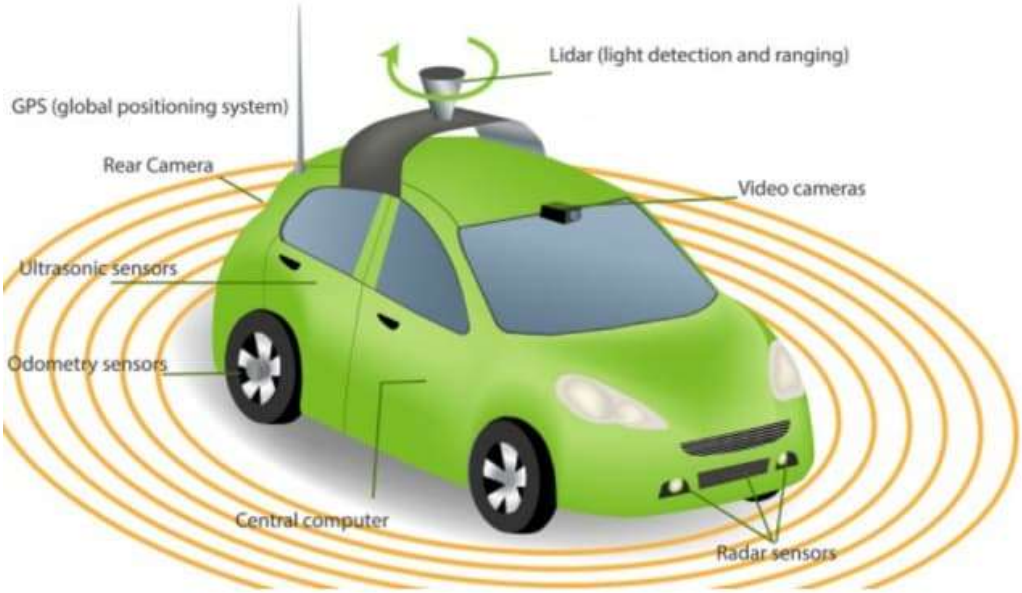
- Basic problem-solving
- Learning through conditioning
- Social behavior (in some species)
- Limited communication

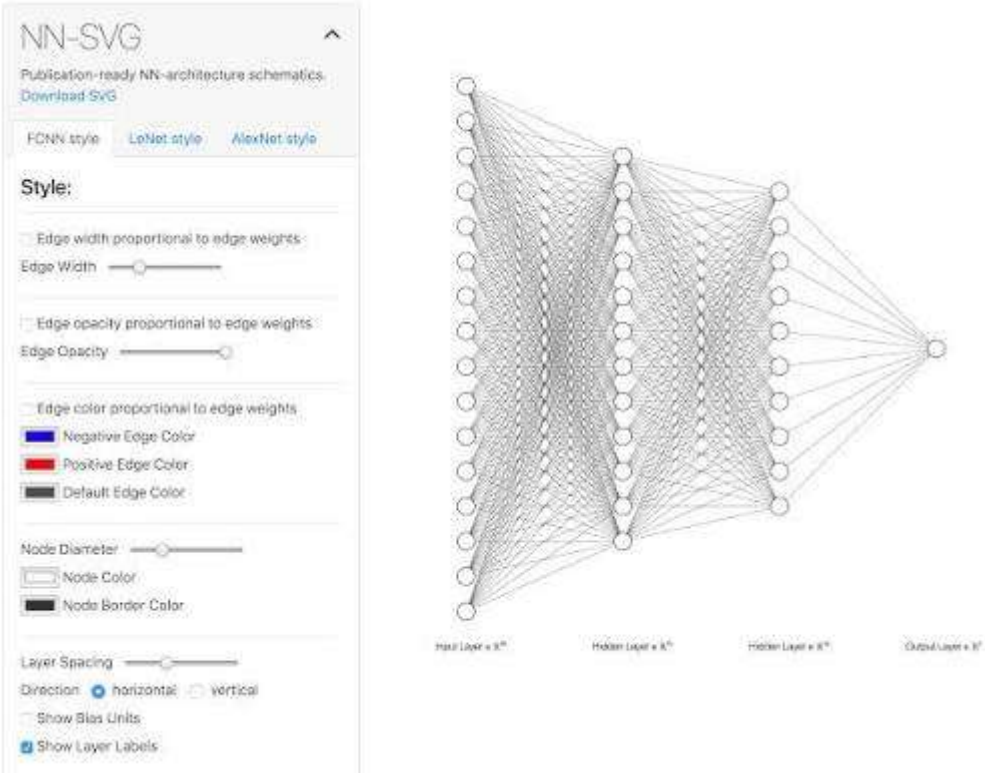
Example

A chimpanzee using a stick to extract termites from a tree.

4. Machine Intelligence




Lidar (light detection and ranging)
GPS (global positioning system)
Rear Camera
Video cameras
Ultrasonic sensors
Odometry sensors
Central computer
Radar sensors



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Definition

Machine intelligence refers to the ability of machines to perform tasks that normally require human intelligence.

It is commonly known as **Artificial Intelligence (AI)**.

Characteristics

- Fast data processing
- Pattern recognition
- Automation
- No emotions
- Learns through algorithms and data

Example

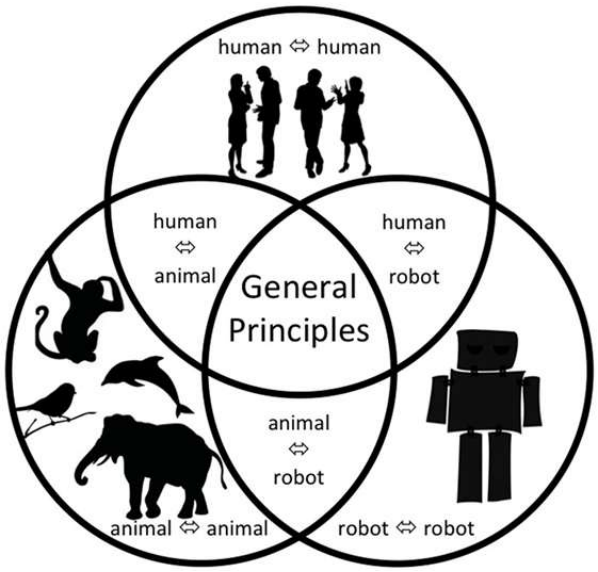
A self-driving car using AI to detect traffic signals and obstacles.

5. Comparison Chart

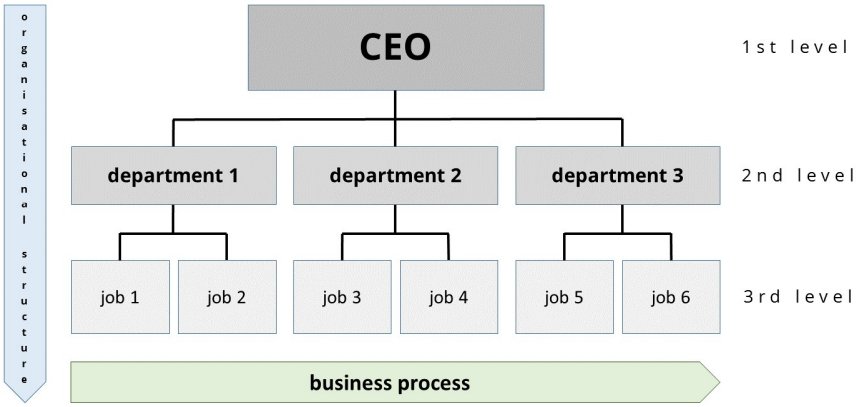
Feature	Human Intelligence	Animal Intelligence	Machine Intelligence
Origin	Natural	Natural	Artificial (Man-made)
Learning Method	Experience, education	Conditioning, instinct	Data & algorithms
Emotional Ability	High	Limited	None
Creativity	Very High	Limited	Very Limited (Pattern-based)
Speed	Moderate	Moderate	Very Fast
Memory	Limited but flexible	Limited	Very Large storage
Adaptability	Highly adaptable	Adaptable in environment	Limited to programming
Self-awareness	Yes	Some species	No

Diagram Representation

(A) Venn Diagram Concept



(B) Hierarchy Diagram



Conclusion

Human, animal, and machine intelligence each have unique strengths.

- **Humans excel in creativity, emotions, and abstract thinking.**
- **Animals excel in survival instincts and environmental adaptation.**
- **Machines excel in speed, accuracy, and handling large amounts of data.**

Understanding these differences helps in developing better AI systems and studying cognitive science.